M2 project goal

Tools and frameworks

Our project uses Tailwind CSS for styling, Vite for build and development tools, both managed through NPM, vanilla HTML and JavaScript. The website is hosted as a Github page and is automatically deployed from the main branch of the project repository.

Core visualizations

Dairy production

To visualise the production of milk and dairy products in Switzerland from 2000 to 2024, we will implement a sankey diagram using D3.js and the d3-sankey plugin/library. The goal is to show the flow of raw milk to different dairy products, in particular the weight losses from processing and making the final product. (How much cheese is produced and what amount of milk is required). The visualization will also include a slider to view different years, so that the user can discover how dairy production has changed over the years in Switzerland. Required lectures: week 4 (D3.js).

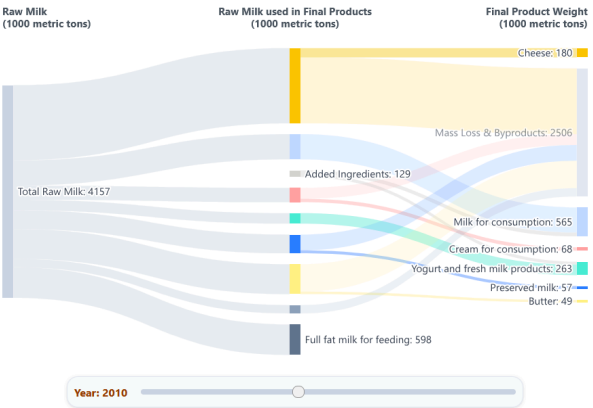

Fig. 1 Our current sankey iteration, found here

The price and production explorer

Since we have a large dataset with many dimensions, it makes sense to visualize them together to uncover hidden relationships. For this purpose, we will develop an interactive, synchronized dashboard. This tool will feature connected charts. As the user navigates through the years, they will observe the growing "economic squeeze" alongside the resulting structural shifts, specifically, the decline in total farm holdings and the simultaneous increase in average herd size. The goal is to make this visualization as customisable as possible: advanced filtering, toggling of elements, chart export, etc… . This setup allows users to directly correlate market pressures with physical farm consolidation over time. This visualization will use standard D3.js.

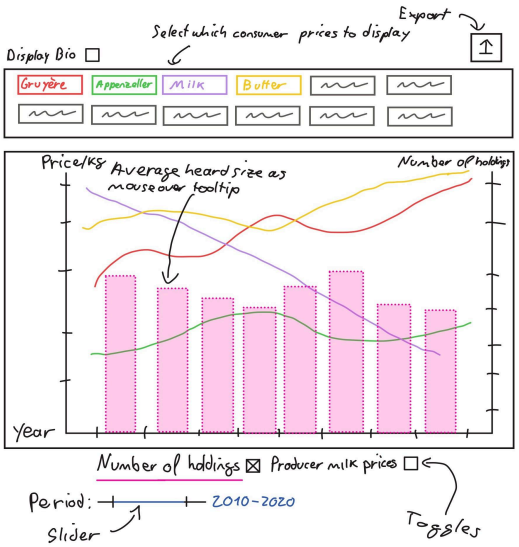

Fig. 2 A sketch of our desired production graph

Regional Cattle distribution

To show how cattle are distributed across Swiss cantons over time, we implement an interactive choropleth map of Switzerland together with a ranked horizontal bar chart for the same year. The data come from the official Identitas AG series on opendata.swiss (Rinder: Entwicklung nach Kantonen); we aggregate monthly values into one total per canton and calendar year. The map and the bar chart stay synchronized: a year slider selects the year for both views, and clicking a canton on the map or a bar highlights the same canton in the other view; clicking the empty map background clears the selection. The implementation uses D3.js and TopoJSON geometries for canton boundaries. Required lectures: week 8 (Practical maps), plus D3 basics from week 4 where relevant for scales and joins.

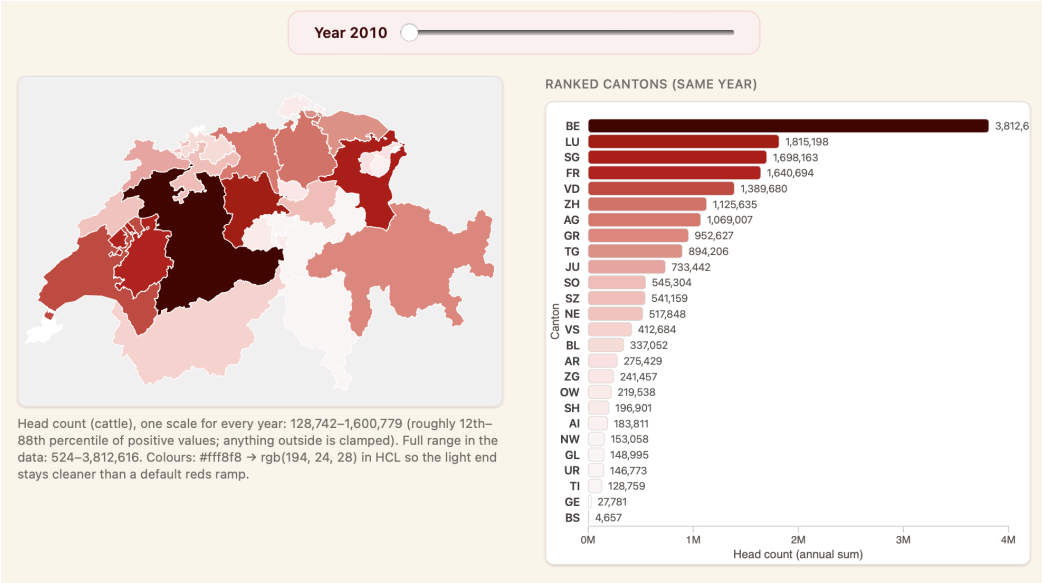

Fig. 3 Current iteration of the canton cattle map and ranking

The consumption per region explorer

As we have explored production and transformation, the next logical step is consumption. From our datasets we have some regional data consumption, which would allow users to detect patterns in regional consumption. For this purpose we propose an interactive set of graphs on an overlaid map of Switzerland. The tool would allow for selection of regions, product types, and date ranges. The goal of the visualisation is to make data as applicable as possible, which requires a focus on visual readability and clarity. This setup should allow users to relate consumer data to real-world insights with ease. The visualisation should also be able to run on D3.js and mostly rely on lecture 8 :Practical maps

Extra ideas

Adopt a cow

At the top of the page, the user could click a button that randomly picks a cow name from the most popular cow names per linguistic region. Based on the linguistic region, the website also picks a corresponding random canton and then displays an image of a cow with text (for example: “Meet Bella from Bern!”). Subsequent explanations and graphs can then use this information to give statistics about the typical cow for that region (for example: “As Bella is from Bern, she has on average 20 other cows in her herd.”), or other visualizations could be set to display default data related to the canton that was picked. This is purely for fun and user engagement and thus would not endanger the project if dropped.